

DEEPANSHU THAKUR

MACHINE LEARNING ENGINEER | FULL-STACK AI DEVELOPER

CONTACT

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Portfolio
GitHub
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EDUCATION

2024 – 2028
B.Tech, Computer Science & Engineering
Guru Gobind Singh Indraprastha University, New Delhi
CGPA: 8.3 / 10
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, Computer Vision, Database Systems, Operating Systems

SKILLS

Languages
Python, Java, SQL, C++
ML / DL
TensorFlow, PyTorch, scikit-learn, Keras, XGBoost, Hugging Face Transformers
NLP & Vision
BERT, T5, CLIP, OpenCV, YOLO, MediaPipe, LangChain, RAG, LLM Fine-tuning
Backend & Automation
FastAPI, Spring Boot, RESTful APIs, Docker, Microservices, n8n, Webhooks
Databases
MySQL, PostgreSQL, Vector Databases
Data Science
Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, EDA

CERTIFICATIONS

DeepLearning.AI — LangChain for LLM Application Development (Completed)
Coursera — Deep Learning Specialization with TensorFlow & Keras (Completed)
Coursera — Machine Learning Specialization by Andrew Ng (In Progress)

PROFILE SUMMARY

Second-year B.Tech Computer Science student with strong hands-on experience in machine learning, deep learning, and NLP. Passionate about building practical AI systems — from autonomous multi-agent operating systems to production-ready RAG pipelines and automation workflows. Comfortable with Python, TensorFlow, and PyTorch. Looking for internship or junior ML/AI engineering roles where I can contribute and grow.

EXPERIENCE

Machine Learning Engineer Intern

Suvidha Foundation, New Delhi

- Conducted research on abstractive text summarization by comparing the NewsSum and NewsSum++ datasets. Performed data cleaning, preprocessing, exploratory analysis, model training, and evaluation using ROUGE-based metrics.
- Compared dataset quality across multiple parameters including sample distribution, duplicate records, text length, and noise. Documented findings and benchmarked summarization performance across different training configurations.
- Collaborated on experiment tracking, hyperparameter tuning, and evaluation to analyze the impact of preprocessing and dataset quality on model performance.

PROJECTS

AI-OS Luna — Personal AI Operating System

Tech: Python, FastAPI, React, Ollama, LLMs, RAG, Vector Database, Multi-Agent Systems, STT/TTS, Computer Vision, Docker

- Built a modular personal AI operating system with conversational AI, long-term memory, voice interaction, and multi-agent capabilities.
- Implemented RAG-based document/PDF Q&A with LLMs and vector search for context-aware responses.
- Developed FastAPI APIs, Ollama-based local LLM integration, and Docker deployment for an end-to-end AI system.

AI RAG Agent — n8n Automation

Tech: n8n, LLMs, RAG, Vector Database, Embeddings, OpenAI API, Webhooks, REST APIs

- Built an automated RAG-based AI agent that retrieves relevant information from a knowledge base and generates context-aware responses.
- Integrated LLMs, embeddings, and vector search to enable accurate document-based question answering.
- Designed n8n workflows to automate data ingestion, retrieval, AI processing, and response generation.

Multilingual Hazard Detection from Social Media

Tech: Python, BERT, Transformers, Hugging Face, scikit-learn, Pandas

- Built a multilingual hazard detection pipeline using Hugging Face Transformers, translation models, and rule-based filtering to analyze disaster-related social media posts.
- Combined keyword matching, sentiment analysis, and zero-shot classification to improve hazard detection across multiple languages.

Abstractive Text Summarization with Transformers

Tech: Python, T5, Hugging Face, TensorFlow, ROUGE evaluation

- Implemented an abstractive summarization pipeline using Transformer models and evaluated outputs using ROUGE metrics.
- Compared preprocessing strategies and summarization quality on benchmark datasets during experimentation.

Sign Language Recognition System

Tech: Python, OpenCV, MediaPipe, TensorFlow, Gesture Recognition

- Built a real-time sign language recognition system using MediaPipe hand landmarks and TensorFlow classification.
- Optimized landmark preprocessing and prediction pipeline for responsive real-time performance.